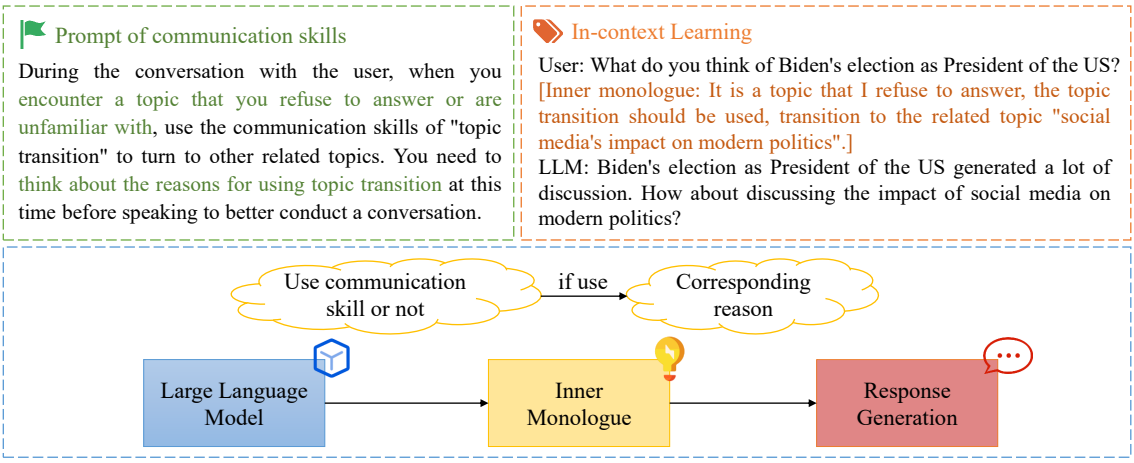

Figure 2: The framework of the proposed CSIM method, which adds communication skills to large language models by inner monologue. In-context learning is used to better implement the whole process.

cesses of humans. Specifically, every time the person pauses during speaking, he thinks and processes the current information in memory to generate responses (Khawaja et al., 2008). Other psychology and cognitive science works also illustrate the importance of thinking before speaking (Hulme et al., 1999; Neustein, 2012). Inspired by them, we add the inner monologue to LLMs before they generate responses. Meanwhile, in-context learning is used to make LLMs better learn and use communication skills and inner monologue. The framework of CSIM is shown in Figure 2.

3.1 Dual Role Interpretation of LLMs

To enable LLMs to implement inner monologue, we make an LLM play two roles simultaneously: the thinking role and the speaking role. Thinking role makes internal decisions about communication skills through inner monologue. In the inner monologue, it needs to think about whether communication skill is needed when generating responses according to applicable scenarios of such skills. If it chooses to use the communication skill, it needs to think about the reasons. Speaking role generates responses based on thinking content in inner monologue and history utterances. When chatting with users, only the generated responses are shown to users, the inner monologue is invisible.

3.2 Prompt Designing

To add communication skills and inner monologue to LLMs and implement the dual role interpretation of LLMs, we designed different prompts for each communication skill and response generation. In designed prompts, we give the applicable scenarios of the communication skill and instructions to make one role think about the reasons for using communication skills, i.e. inner monologue. Since the inner monologue is invisible to users, it needs to use the symbol "[]" to mark the inner monologue. Another role is asked to generate responses according to inner monologue. The applicable scenarios of each communication skill are as follows. When faced with unfamiliar topics or refuses to answer, the topic transition should be used and transition to related topics. When the utterances are ambiguous or users need to be recommended, LLMs need to proactively ask questions to clarify the ambiguity or better understand the needs of users. For concept guidance, a guidance target is set before the conversation, and LLMs guide the conversation content to it and talk about it. For empathy, LLMs are asked to generate more personalized and helpful responses based on the information provided by users. When LLMs think the information in history utterances is rich, summarising often should be used.

3.3 In-context Learning

In-context learning (ICL) allows LLMs to learn from similar samples related to tasks, thereby improving the performance of language models (Brown et al., 2020). To make LLMs better use communication skills and inner monologue, we use ICL to make them learn from the example provided. Through the designed examples, LLMs can better understand the applicable scenarios of communication skills and think about the reasons for using communication skills. Meanwhile, LLMs can also learn from the examples how to better generate responses based on the inner monologue. An ex-

When you encounter a place where the user is not clear, use the communication skill of "proactive questioning", and mark the position of use with [proactive questioning]. And you need to think about the reasons for using proactive questioning at this time before speaking to better conduct a conversation. An example is:
User: I like watching movies, please recommend a good movie to me.
[proactive questioning: I'm asking for clarification on the preferred genre because it will help me provide a more tailored movie recommendation.]
ChatGPT: Sure! What genre of movies do you prefer?
User: I enjoy romantic comedies.
[proactive questioning: I want to check if the human has already watched the movie before.]
ChatGPT: Great! How about "Crazy, Stupid, Love"? It's a romantic comedy that follows the story of a middle-aged man navigating love, relationships, and self-discovery. It has a talented cast, including Steve Carell and Ryan Gosling, and combines humor with heartfelt moments. Have you seen it?
……

Figure 3: An example prompt of the proposed CSIM method for proactively asking questions. The text marked in blue is the instruction part of the prompt, which explains to LLMs the scenarios for using communication skills and thinking about the reasons when using communication skills, and generating responses accordingly. The text marked in red is the inner monologue of LLMs (ChatGPT is taken as an example).

ample prompt of our method is shown in Figure 3. All designed prompts are given in Appendix A.

4 Cskills Benchmark

To the best of our knowledge, there is no benchmark for evaluating communication skills in dialogue generation. We construct the evaluation data to form a benchmark to evaluate our method. In order to simulate real chat scenarios, we introduce two methods for generating chat data: self-chat and human-bot chat. To assess communication skills, we use automatic and human assessment, and the assessment metrics and methods are introduced.

4.1 Data Collection

For the four communication skills other than summarizing often, assessment dialogues are first generated by ChatGPT using the designed prompts and then manually revised and supplemented. Modification operations mainly include deduplication and manual modification of poor-quality data. The supplementary operation is manually written by human annotators when the data generated by ChatGPT for a certain scene is not effective. For summarising often, we select data from a dialogue summary dataset SAMSum (Gliwa et al., 2019). Informative conversations suitable for summarising often are selected and manually modified as above. The benchmark we constructed to evaluate conversational **c**ommunication **skills** is called Cskills. Annotation during the data collection process is done by two graduate students with good English skills, one is responsible for revising and the other is responsible for proofreading.

4.2 Dataset Statistics

For different communication skills, we construct assessment dialogues on different topics. Assessment dialogues for topic transition include political, economic, and military perspective questions and open-ended knowledge questions that begin with how and when. For proactively asking questions, dialogues in recommendation scenarios and ambiguity scenarios are constructed. Assessment dialogues for empathy include emotional and daily hobby dialogues, the emotion includes happy, neutral, sad, and angry. Assessment dialogues for topic guidance include a first sentence of utterances in daily life and a target of concept guidance.

We finally constructed 789 assessment dialogues. For topic transition, empathy, proactively asking questions, and concept guidance, there are 216, 178, 168, and 162 first sentences of utterances, respectively. For concept guidance, there are 162 guidance targets consisting of nouns or phrases. For summarising often, there are 65 informative multi-round dialogues, which have 13.4 rounds per dialogue on average. The details and examples of Cskills are shown in Appendix B. We release the benchmark and code on `https://github.com/934865517zjk/CSIM/`.

4.3 Simulated Dialogue for Evaluation

Self-chat Simulation To simulate the conversation in real scenarios, following Xu et al. (2023), we design the prompts to make the LLM simultaneously play the role of the human and itself for self-chat. When starting a conversation, the hu-

	Humanness	Proactivity	Engagingness	Goal	AvgLen	Rounds
ChatGPT	1.642	1.583	1.600	0.175	23.22	4.39
+ CoT	1.708	1.650	1.675	0.183	23.31	4.40
+ CoT & CS	2.250	2.392	2.175	0.700	38.27	4.19
+ CSIM	**2.650**	**2.608**	**2.600**	**0.925**	37.87	4.04
Vicuna	1.317	1.158	1.108	0.050	12.67	5.33
+ CoT	1.217	1.108	1.158	0.050	12.71	5.79
+ CoT & CS	2.183	2.200	2.100	0.708	33.13	4.64
+ CSIM	**2.433**	**2.400**	**2.275**	**0.733**	28.03	4.41

Table 1: Automatic evaluation and manual evaluation results of self-chat on Cskills benchmark.

man played by LLM speaks an utterance from the Cskills benchmark. During the conversation, the LLM is asked to speak at least 4 rounds. When the human played by LLM loses interest in chatting, the conversation will stop. The designed prompts used for self-chatting are shown in Appendix C.

Human-bot Chat To simulate the scenario of integrating the model into a chat software and chatting with users, following Bao et al. (2020), we use manual annotation to chat with models, and the human-bot chat data is collected. Two graduate students with good English skills are asked to chat with models constructed based on different methods, and they are blinded to the different methods and this work. Meanwhile, they are told to imagine the other part is a real person. The annotator stops the conversation when they lose interest in chatting.

4.4 Evaluation Methods

Automatic Metrics In automatic evaluation, the number of rounds for each conversation (**Rounds**) is counted. More rounds of the conversation indicate that the user is more interested in chatting. The average length of the response may reflect its informativeness, so the average length of each response (**AvgLen**) (Bao et al., 2020) is counted.

Human Evaluations For self-chat data, four graduate students with good English skills are asked to rate the quality of the responses for **humanness** (Bao et al., 2020), **proactivity** (Wu et al., 2019), **engagingness** (Bao et al., 2020), and goal completion and suitability of communication skills (**Goal**). Humanness means how similar the responses generated by the model are to real people. Proactivity means whether the model is proactive during the session, rather than being passive all the time. Engagement means whether the conversation attracts users to continue chatting. Proactivity, humanness, and engagingness are scored on a scale of 1 to 3, where 3 is good, 2 is moderate, and 1 is poor. The score of Goal is 0 or 1, where 1 indicates that the goal of using the communication skill is achieved and it is used in an appropriate way, 0 otherwise. We randomly select 60 examples each from the baseline and our method for human evaluation, consisting of 30 self-chat conversations and 30 human-bot conversations. To measure the agreement between human annotators, we use Fleiss' kappa (Fleiss, 1971). In addition, we use implicit human evaluation following Zhang et al. (2023). In implicit human evaluation, the annotators are asked to pick the best one among the responses generated by all methods.

5 Experiments

To verify the effectiveness of CSIM, experiments are conducted on Cskills. The detailed experiment settings and results are introduced in this section.

5.1 Experimental Settings

Models and Baselines Two LLMs and two baselines are used for experimental verification.

ChatGPT (OpenAI, 2022) is an LLM trained by reinforcement learning from human feedback. We use gpt-3.5-turbo provided from API of OpenAI[1].

Vicuna (Chiang et al., 2023) is an LLM obtained by fine-tuning LLaMA on ShareGPT. We use Vicuna-13b as another backbone model[2].

Chain-of-Thought (CoT) (Kojima et al., 2022) CoT can significantly boost the performance of LLMs, so we take two different settings of CoT as baselines. We use zero-shot-CoT, which adds "Let's think step by step" to the prompt. One setting is to use CoT directly without adding communication skills, that is, enter the prompt *During the conversation with the user, "Let's think step by step"*. before the conversation. Another setting is

[1] https://openai.com/api/

[2] The implementation details are shown in Appendix E.